Pre-analysis Plan

Sample selection criteria:

The analysis focuses on the sample with at least 1 child, everyone without children is excluded from the analysis sample.

H1. Having daughters leads to stronger support for gender equality (as measured by attiitudes). That is, denoting X as a measure of support for gender equality, (X for respondents with daughters) > (X for respondents without daughters).

Outcome variable:

I measure gender attitudes using the 5 questions that are available for the year 2019: *plh0395i01, plh0395i02, plh0395i03, plh0395i04, plh0395i05, plh0395i06*

I clean the variables by recording everything outside {1, 7} as missing, and not part of the analysis of this hypothesis.

I then create 1 factor that measures gender attitudes, that is driven by the correlations in the data. I specifically use confirmatory factor analysis, sample code:

factor *plh0395i01 plh0395i02 plh0395i03 plh0395i04 plh0395i05 plh0395i06* factors(1) // Extract 1 factor

Variable of interest:

I use the variable *biobirth/kidsex01* to measure the first child's gender. I recode this variable as equal to 1 if the child is male, 0 is female, missing is the variable has any other value.

Since other possible children in the household are endogenous (number as well as timing), the full gender composition of the household's children is not of key interest.

Controls:

Parent sex: I use the variable: *ppathl/sex* to determine respondent's sex. Variables is recodes as equal to 1 for male, equal to 0 for female, missing for all other values of the variable.

Number of children: *hl/hlc0043: Number Children* is added as the number of children, other values of the variable are excluded/ used as missing.

Regression model is an OLS:

Gender_attitude = sex_1st_child parent_sex Nr_of_children

Sample:

I only include people who have at least one child (everyone without children is excluded from the analysis). Furthermore, since gender attitudes are only measured in 2019, I will use this data, not other years. If it turns out other variables are not measured in 2019(or participants didn't fill in the survey in 2019) I chose their data from the most recent year prior to 2019.

Interpretation of the results:

A p-value <0.05 for the sex_1st_child coefficient would be in line with the hypothesis.

H2. Having daughters leads to more left-leaning political preferences. That is, denoting Y as a measure of left-leaning political preferences, (Y for respondents with daughters) > (Y for respondents without daughters).

I use the exact same sample and specification as for the test of H1 (above). Only the outcomes variable is different. I create a binary variable the equals 1 for ' left-leaning political preferences, and 0 for others. I base the categorization on Germany: Political parties at a glance. The recoded variable is: *plh0333: Partei Bundestagswahl* and equals 1 for categories: Linke, SPD, Piraten, Tier, Grune and equals 0 for: FDP, CDU/CSU, AfD

The regression model is then: left-leaning_pol = sex_1st_child parent_sex Nr_of_children,

Where I find evidence in line with the hypothesis if sex_1st_child (daughter) has a p<0.05.

Sample period: The most recent wave in which the left-leaning_pol (plh0333) is measured is 2018. Therefore here I use the data from wave 2018.

H3. Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes). That is, denoting Z as a measure of gender equality in the household, (Z for respondents with daughters) > (Z for respondents without daughters).

I operationalize gender equality in the household by time use. Specifically I will create an ' average number of hours per day on housework' as the measure and compare parents with first born daughters and sons, in the same specification as before:

The time use chores variable is created as follows: avg_hours_chores *(bjp_05_03*5+* bjp/bjp_05_23 + bjp/bjp_05_13) /7

The time use children variable is created as follows: avg_hours_child= (*bjp/bjp_05_04**5 + *bjp/bjp_05_24* + bjp/bjp_05_14) /7

And the sum of child and chores hours

For this I also use the sample of the year 2019

Regression model, OLS:

I estimate 3 regressions, specifically:

avg_hours_chores = sex_1st_child parent_sex Nr_of_children

avg_hours_child = sex_1st_child parent_sex Nr_of_children

avg_hours_sum= sex_1st_child parent_sex Nr_of_children

The variable of interest is sex_1st_child. We find evidence in line with the hypothesis if parents with first born daughters have a significant higher average number of hours ($p<0.05$) for each regression.

H4. Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes). That is, denoting X as a measure of support for gender equality, X for male respondents with daughters – X for male respondents without daughters! > X for female respondents with daughters – X for female respondents without daughters!

Same as hypothesis 1, but than with an interaction for gender, specifically:

Gender_attitude = sex_1st_child parent_sex sex_1st_child*parent_sex Nr_of_children

If the interaction term *parent_sex sex_1st_child*parent_sex* is significant (i.e. father*1st born daughter), we find evidence in line with our hypothesis.